



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 10	Team No-	Project Title- Disaster Alert & Community Response App
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1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The Disaster Alert & Community Response App is a centralized platform designed to improve communication and coordination during disasters such as floods, cyclones, earthquakes, fires, and other emergencies. During such situations, people may face difficulties in receiving timely alerts, reporting incidents, finding nearby assistance, and communicating with response teams. The proposed application provides a digital platform where users can receive disaster alerts, report emergency incidents, request assistance, and offer help to affected people. The system maintains user, disaster, incident, and response information using suitable database technologies. A distributed backend architecture with REST APIs enables efficient communication between the application and backend services. Authentication and authorization can be implemented to protect user and emergency-related information. The system can use separate services for users, disaster alerts, incident management, and community responses, making the application scalable and maintainable. By providing a centralized platform for information sharing and community participation, the proposed system aims to support faster reporting, better coordination, and more effective community response during emergencies.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090017	L. Tejasri	L. Tejasri
2520090235	Rohita	Rohita
2520090007	S. Dikshita	S. Dikshita

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- Frontend: Not specified in the handout
- Backend: FastAPI
- Databases: PostgreSQL, MongoDB
- API: REST API, OpenAPI
- Authentication: JWT Authentication
- Backend Services: Node.js, Express.js, Spring Boot
- Database Libraries: SQLAlchemy, asyncpg, Motor, Mongoose, Spring Data JPA
- Validation: Pydantic
- Messaging: Apache Kafka
- Testing: Pytest, httpx
- Containerization: Docker
- CI/CD: GitHub Actions
- Version Control / Repository: GitHub
- Deployment: Docker Compose, GitHub Container Registry (GHCR)

Guide Name and Signature (remarks if any)

A handwritten signature in blue ink, appearing to be 'Duy Nguyen', written over a horizontal line.